Visualization Framework for Shared Emotional Awareness in Personalized VR for Stroke Recovery

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***Abstract*—Stroke affects millions of people worldwide and often leads to long-term motor, cognitive, and social impairments requiring sustained rehabilitation. However, rehabilitation programs frequently rely on standardized approaches, despite the diverse needs of individual stroke survivors. Emotional state is particularly relevant in this context, as it influences engagement, motivation, and interaction during therapeutic activities. This work proposes EMOCAP, a visualization framework for shared emotional awareness, enabling the personalization of Virtual Reality (VR) environments, tailored to stroke rehabilitation. The framework, created using a Human-Centered Design (HCD) methodology with members of a rehabilitation center, allows to capture, process, distribute, and visually represent emotional states during rehabilitation sessions. It integrates an (client-side) emotion recognition module, a (server-based) communication layer for sharing affective information across connected users, and a VR module enabling presentation of virtual stimuli, while a visualization component externalizes emotional states through lightweight symbolic representations. Results from an initial evaluation with 16 participants are presented and discussed, showcasing the potential of the framework to support interaction among stroke survivors, healthcare professionals, and caregivers, while establishing emotional awareness as a foundational layer for future personalization strategies in VR-based rehabilitation.**

***Index Terms*—Emotion Visualization, Affective Computing, Virtual Reality, Personalization, Stroke Rehabilitation**

I. Introduction

Stroke is a leading cause of long-term disability worldwide, resulting from an interruption of blood flow to the brain that can cause significant physical, cognitive, and emotional impairments [1], [2]. Rehabilitation is a critical part of recovery, aiming to help stroke survivors regain lost functions, improve their independence, and enhance their overall quality of life. Through targeted exercises, therapy, and support, rehabilitation promotes neuroplasticity, i.e., the ability of the brain to reorganize and adapt, allowing individuals to recover their abilities and adapt to new challenges over time [3], [4].

Regardless of its importance, traditional rehabilitation methods tend to follow a one-size-fits-all approach, applying uniform practices to all survivors without recognizing the unique nature of each person's condition [5], [6]. These strategies often fail to consider the wide range of differences between individuals, including the severity and location of the brain injury, physical and cognitive abilities, emotional responses, social support, or cultural background. As a result, interventions may not fully align with the specific needs of a survivor, which can limit engagement and slow progress.

To achieve more meaningful outcomes, rehabilitation must be flexible and personalized, evolving with each survivor's journey to address their distinct capabilities [7]–[9]. A key element for improving personalization is the emotional state of stroke survivors, as emotions constitute complex, multidimensional responses that reflect an individual's internal state and significantly influence behavior, motivation, and interaction [10]–[12]. In the context of stroke rehabilitation, understanding these emotional signals can be particularly valuable, as survivors often experience a range of affective states that impact their engagement and recovery [13], [14]. In stroke survivors, where verbal communication or conscious self-reporting may be limited, emotion recognition becomes even more critical for guiding responsive and empathetic care [15], [16]. Here is where visualizing the emotional state becomes paramount, allowing latent affective signals to be transformed into accessible visual representations that support real-time interpretation, shared emotional awareness, and more informed therapeutic interaction.

In this vein, this work proposes EMOCAP, a visualization framework for shared emotional awareness in personalized Virtual Reality (VR) for stroke rehabilitation. The framework performs emotion recognition and broadcasts the corresponding affective states across a shared VR experience, where emotions are presented through lightweight symbolic representations that support immediate awareness and interpretation. In this way, the framework establishes visual emotional awareness as a foundational layer for the personalization of the VR experience, enabling emotional shifts to be identified and considered when adapting therapeutic activities. This emotion-aware approach seeks to enhance user engagement, foster

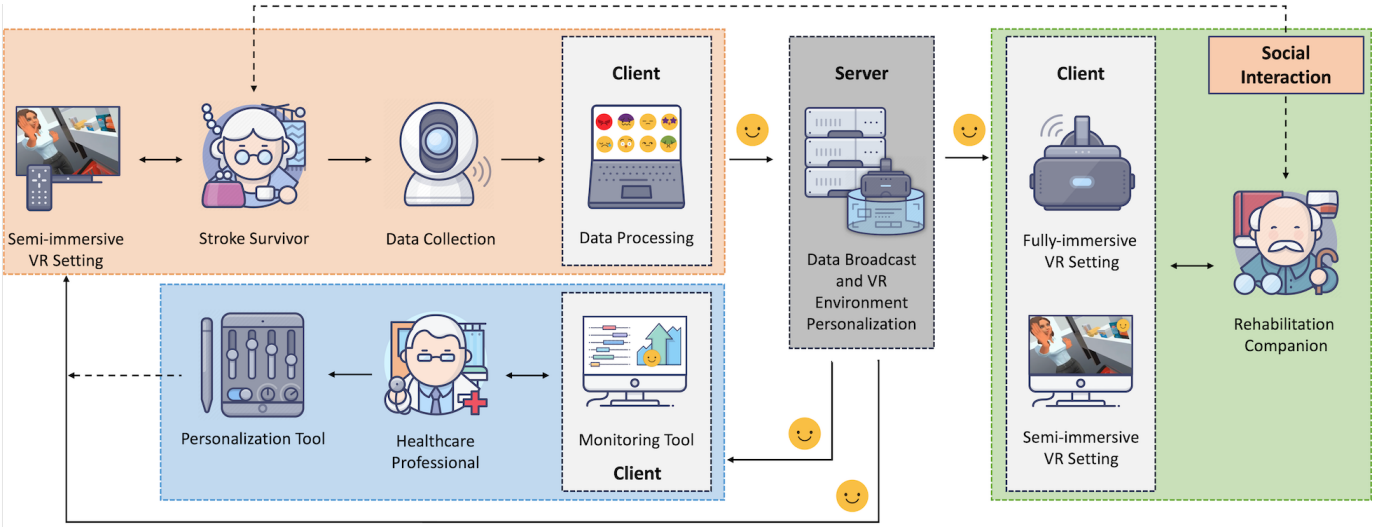


Fig. 1. Conceptual model showing how the EMOCAP framework performs emotional data acquisition, affective state processing, server-mediated distribution, and visual representation of shared emotional awareness within personalized VR environments for stroke rehabilitation. Inspired by the work of [5].

empathy, and support more effective and motivating recovery processes. Additionally, healthcare professionals can monitor emotional evolution over time, gaining insight into how users respond to the VR intervention and enabling informed adjustments whenever needed, thus contributing to a more responsive and adaptive rehabilitation experience.

The remnant of this paper is structured as follows. Section 2 delves into the methodology, main features of the proposed framework, and description of the tools used. Next, Section 3 reports the results from an initial study with 16 participants and discusses the main insights. Last, conclusions and future directions are drawn in Section 4

II. Methods and Materials

This section describes the methods and materials used to create the framework (Figure 1).

A. Methodology

The design and development were guided by a Human-Centered Design (HCD) methodology (Figure 2), focusing on the needs, preferences, and lived experiences of both the target audience and domain experts, ensuring that the resulting framework effectively addresses real-world rehabilitation challenges. This approach emerged from a collaborative process that included healthcare professionals and stroke survivors from a leading rehabilitation center, alongside academic specialists in VR, Human–Computer Interaction (HCI), and Computer-Supported Cooperative Work (CSCW).

B. EMOCAP Framework for shared Emotional Awareness

The conceptual model illustrated in Figure 1 provides a high-level overview of the EMOCAP framework for shared emotional awareness, structured around three interconnected components linked through a client–server architecture: (i) emotional data acquisition and classification, (ii) visual dissemination of affective state across rehabilitation participants,

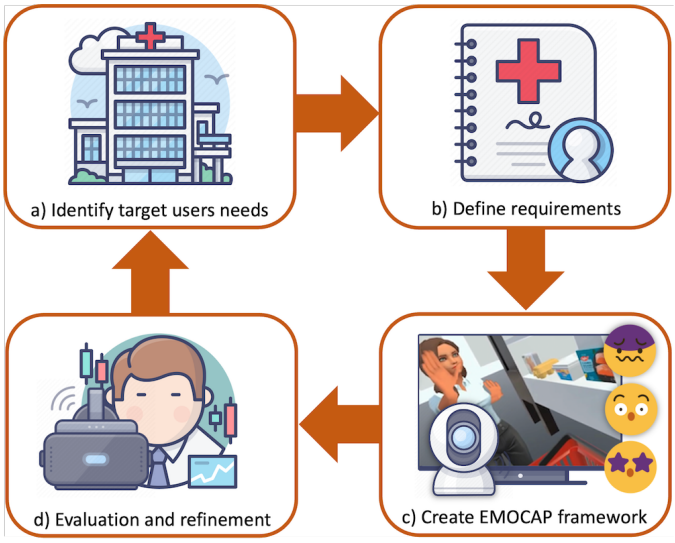


Fig. 2. Methodology adopted for the design of a visualization framework for shared emotional awareness: a) identification of target users' needs; b) definition of requirements; c) creation of the framework; d) user evaluation and refinement.

and (iii) clinical monitoring and intervention support. Together, these components establish a visualization pipeline in which emotional states are captured, processed, distributed, and represented in real time to support social interaction and adaptive rehabilitation within personalized VR environments.

The orange region represents the initial stage of the framework, where a stroke survivor engages in a semi-immersive VR setting while emotional data are continuously collected through a camera-based sensing module (Figure 3). Captured facial cues are processed locally at the client side, enabling real-time emotion classification before transmission to the central server. This stage constitutes the starting point of

the emotional visualization broadcast pipeline, transforming affective signals into structured information that can later be shared with other actors involved in the rehabilitation process.

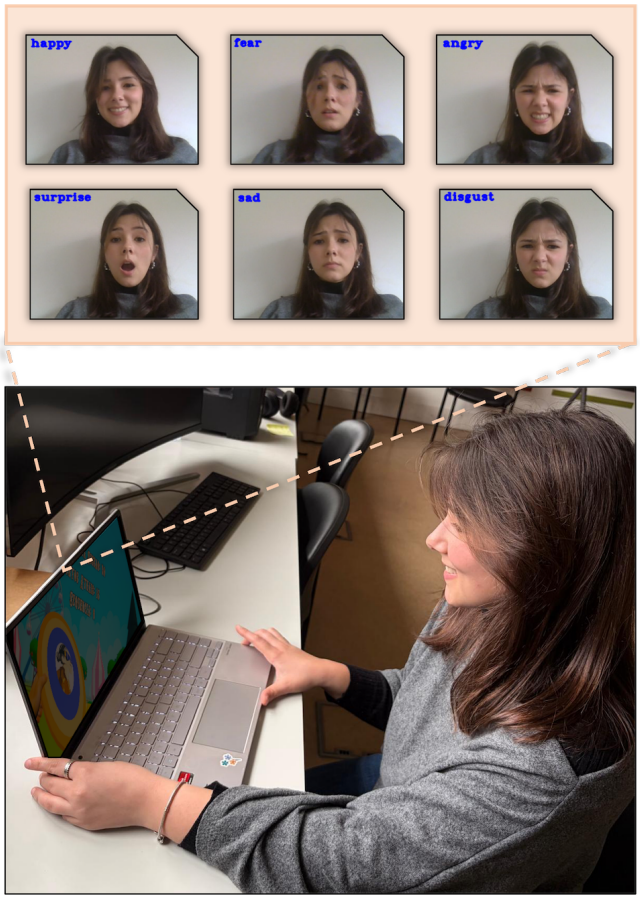


Fig. 3. Emotion recognition component of the framework during data acquisition, showing a participant interacting with the semi-immersive setup while producing facial expressions associated with distinct emotional classes recognized by the framework.

In parallel with emotional classification, the server includes an automatic personalization mechanism capable of triggering adaptive modifications to the VR environment according to the detected emotional state. These changes are then disseminated back to connected clients, tailoring the rehabilitation experience to the survivor's current affective condition.

To operationalize this adaptive process, the VR rehabilitation module adopts a country fair environment [17], where therapeutic activities are embedded within playful virtual tents. Each area metaphorically reproduces elements of a traditional fairground while supporting rehabilitation-oriented motor and cognitive exercises. Within this environment, emotional feedback can influence the presented stimuli. For example, when sadness is detected over a prolonged period, the framework may introduce more positively valenced audio, reduce task difficulty, or adjust environmental stimuli to promote engagement and emotional recovery. Each serious game can also be progressively adapted according to the survivor's functional abilities. In a ball-toss activity, parameters such as target size, target quantity, throwing distance, and object dynamics may be adjusted to control task difficulty. Early sessions may use larger and closer targets, while later stages progressively increase challenge by reducing target dimensions and extending target distance.

The green region represents the visual dissemination of emotional state to another participant involved in the rehabilitation session, who may interact through either a semi-immersive or fully immersive VR setting. Emotional information received from the server is displayed through emoji-based visual encoding, allowing affective states to be perceived immediately and non-intrusively during interaction (Figure 4). This visual layer supports shared emotional awareness by externalizing otherwise latent affective information, enabling rehabilitation companions to better understand the survivor's emotional condition. By making emotional states visible in real time, EMOCAP strengthens social presence and promotes empathetic interaction, which may contribute to engagement and motivation during rehabilitation sessions.

The blue region corresponds to the clinical monitoring layer, where healthcare professionals access a dedicated monitoring client connected to the same server infrastructure. Through this interface, professionals can follow the survivor's emotional progression, observe finished VR sessions (Figure 4), and review affective patterns alongside other rehabilitation performance indicators. Emotional states received from the central server are presented not only as immediate status information, but also as structured temporal records that support interpretation beyond the active rehabilitation session (Figure 5). In addition to passive observation, this component enables manual intervention through a personalization tool, allowing professionals to suggest or apply targeted modifications to the VR setting when necessary. Such interventions may complement the automatic personalization module by incorporating clinical judgment into the adaptive rehabilitation process.

In summary, through this three-part organization, EMOCAP positions emotional state not only as a measurable signal, but as a visual and actionable element capable of supporting adaptive rehabilitation, social awareness, and clinical decision-making within personalized VR environments.

C. Tools used

Regarding the emotion data collection, classification, and broadcast components of the framework, they were implemented primarily in Python, which served as the main development language for integrating all system modules. Real-time facial emotion recognition was supported by FER, while OpenCV was used to capture webcam input and process image frames before analysis. Face detection and localization were performed using MTCNN to ensure that facial regions were identified prior to emotion classification. The graphical interface was developed with CustomTkinter, with support from Pillow for image handling within the interface. Emotion inference relied indirectly on deep learning models provided through TensorFlow and Keras via the FER library. Emotional states were visually represented using Unicode emo-

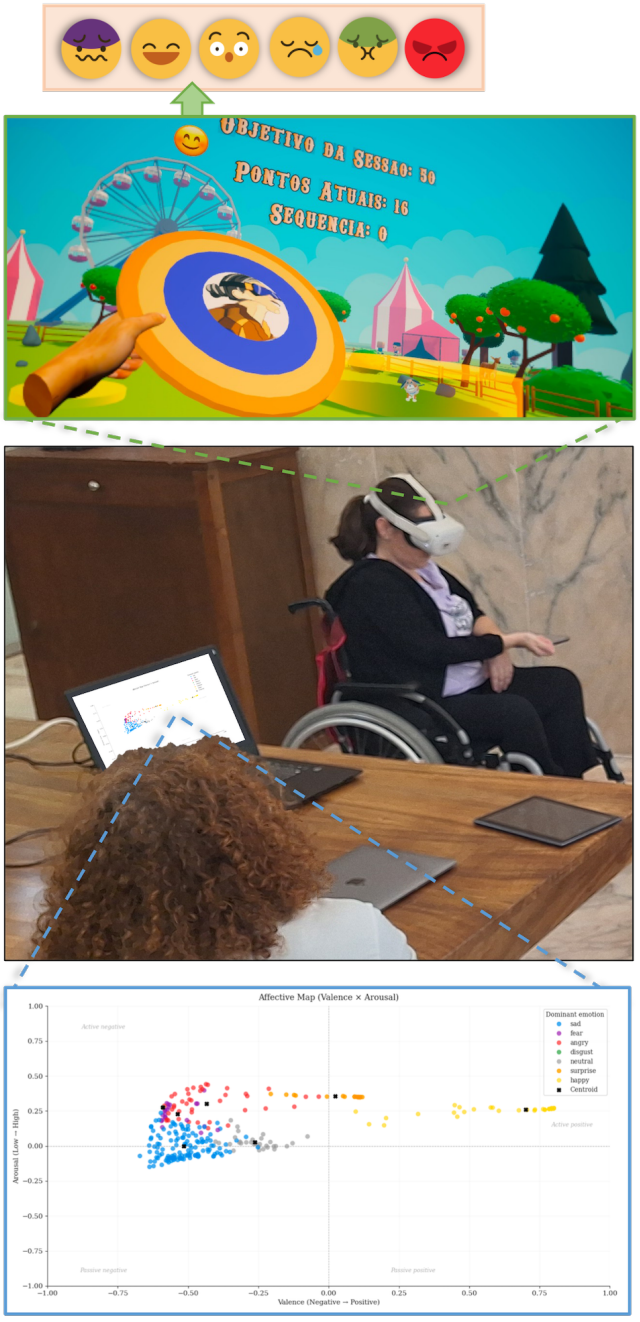


Fig. 4. Visualization of shared emotional awareness in the immersive VR setting, where a participant perceives another individual's emotional state through emoji-based visual encoding, while a healthcare professional monitors emotional progression through complementary visual analytics.

jis, enabling lightweight symbolic encoding without external graphical packages. Moreover, socket-based communication supported the client–server architecture responsible for broadcasting emotional states across connected users.

As for the VR environment, it was deployed on the Meta Quest 3 and implemented using the Unity game engine, with interaction scripts developed in C#. A post-process effect with increased contrast and saturation was applied to reinforce the carnival-inspired aesthetic, while the heads-up display adopted



Fig. 5. Monitoring view showing end-of-session emotional quantification through a bar chart, together with a timeline that allows healthcare professionals to examine how emotions evolved throughout the rehabilitation session.

the Carnevalee FreakShow font to maintain visual coherence with the environment. The scene integrates original 3D assets modeled in Blender, including elements such as trees, bushes, the dog, bow, and frisbee, together with complementary low-poly assets obtained from Poly Pizza under open licensing, including circus tents, pedestrian characters, and a Ferris wheel. Spatial audio was implemented through FMOD Studio to provide directional auditory cues and dynamic sound adaptation, including pitch and volume variation to reduce repetition and strengthen immersion.

III. Initial Assessment and Discussion

To evaluate the emotion recognition approach, as well as the visual representation within a VR setting, a user study was conducted. Participants were instructed to use the framework on a dedicated laptop computer, where they were exposed to virtual stimuli presented in a semi-immersive setting. During this process, their emotional state was captured and displayed in real time through the proposed visualization interface. Participants were also given the opportunity to experience the VR environment in which the stimuli were presented, allowing them to observe the emoji-based emotional representations integrated into the virtual setting. Finally, they explored the monitoring features of the framework, which enabled them to examine how their emotional state evolved over time. A researcher remained present throughout the session, providing assistance when necessary and documenting relevant observations during the interaction.

The participants sample had a mean age of 21.44 years (N=16), comprising 56.25% female participants (9). Regarding prior experience with VR, 68.75% of participants (11) reported previous use of VR for different areas of application, while 31.25% (5) had no prior experience. Participants were first introduced to the objectives, tasks, and overall experimental setup and design. After providing informed consent, they filled a questionnaire regarding their background (e.g., demographic data, background, experience with VR technologies). Next, participants familiarize themselves with the features of the framework. Upon completing the assigned tasks, they filled out a post-study questionnaire covering their experiences, preferences and possible suggestions. The study concluded with a follow-up interview to collect qualitative feedback.

The study was conducted in a dedicated virtual and augmented reality laboratory space with controlled lighting and minimal noise to create an optimal, distraction-free environment. Data were collected at the end of the experience using a post-task questionnaire that combined a Likert-type scale (ranging from 1 – Low to 5 – High) with open-ended questions to capture both quantitative ratings and qualitative feedback. The survey focused on several key dimensions: D1 - *The system accurately recognizes my emotional state*; D2 - *The emotion feedback felt responsive in real time*; D3 - *Changes in the displayed emotion were easy to follow*; D4 - *I understood what the displayed emotion label represented*; D5 - *I feel comfortable having emotions monitored*; D6 - *Seeing emotional state increases my trust in the system*; D7 - *Seeing emotional state supports my motivation and engagement*. In addition, participants' preferences and impressions were further explored through a post-study interview.

A. Results

The System Usability Scale (SUS) results indicate a positive usability perception, with an average score of 87.3, which is substantially above the accepted benchmark of 68. This score suggests that participants found the framework easy to understand, intuitive to use, and suitable for interaction with minimal effort. Furthermore, it suggests that the emotional awareness component was successfully integrated without introducing significant cognitive burden, allowing participants to interpret emotional feedback naturally during interaction.

Figure 6 provides an overview of the findings using both a box-plot and a radar-chart representation to illustrate the different dimensions considered. Regarding D1, the overall median score was 4, suggesting that participants generally considered the displayed emotional output to be consistent with their perceived affective state, supporting the credibility of the visual feedback. As to D2, the overall median score was 5, indicating that participants perceived the emotional visualization as immediate, an important factor for maintaining temporal coherence between expression and feedback. For D3, the overall median score was 4.5, suggesting that emotional transitions were visually understandable and could be followed without major difficulty. In terms of D4, the overall median score was 5, indicating that the emoji representations and other visual cues were clear and easily interpretable during interaction. On the subject of D5, the overall median score was 4.5, suggesting general acceptance of emotional monitoring within the proposed framework. With respect to D6, the overall median score was 4.5, indicating that visualizing emotional state contributed positively to the participants experience while using the framework. Finally, for D7, the overall median score was 5, suggesting that emotional visualization was perceived as a motivating element. All in all, these insights seem to indicate the potential of lightweight visual cues as an effective mechanism for externalizing affective information within VR-based stroke rehabilitation environments.

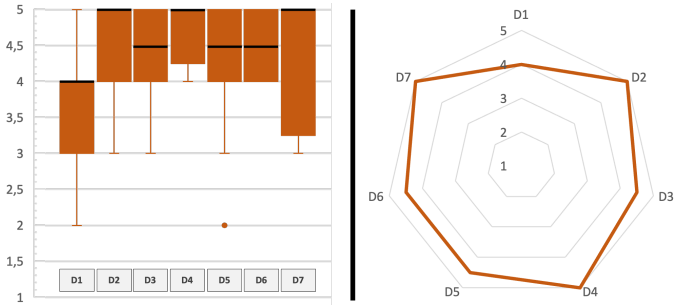


Fig. 6. Box Plot (left) and Radar Chart (right) representations, illustrating the overall experience using a Likert-type scale: 1- Low; 5- High for the dimensions considered: D1 - *The system accurately recognizes my emotional state*; D2 - *The emotion feedback felt responsive in real time*; D3 - *Changes in the displayed emotion were easy to follow*; D4 - *I understood what the displayed emotion label represented*; D5 - *I feel comfortable having emotions monitored*; D6 - *Seeing emotional state increases my trust in the system*; D7 - *Seeing emotional state supports my motivation and engagement*.

Regarding participants' comments and suggestions, the emoji-based representation was perceived positively, mainly due to its immediate recognizability and low interpretative effort. Participants considered this lightweight symbolic encoding effective for communicating emotional states in real time. Still, some participants suggested complementing the current approach with additional visual cues, such as color-based indicators or 3D avatar expressions, to further enrich emotional interpretation and improve perceptual expressiveness. Concerning the monitoring capabilities, the timeline feature was particularly valued for providing a clear retrospective overview of emotional evolution throughout the experience, while the emotion distribution summary was recognized as useful for identifying which affective states were most prevalent during interaction. Participants also suggested integrating synchronized video recording with the timeline, allowing emotional fluctuations to be examined alongside recorded behavior, so that healthcare professionals could verify what survivors were doing and how they were reacting at specific moments when interacting with the temporal visualization. This could also be complemented with a recording of the VR experience, enabling emotional events represented in the timeline to be examined together with the corresponding virtual interaction context, thereby allowing professionals to better understand how specific tasks, stimuli, or in-game situations influenced survivors' emotional responses.

IV. Concluding Remarks and Next Steps

The emotional state of stroke survivors plays an important role in their rehabilitation, as it affects motivation, attention, and willingness to continue the recovery process. In VR-based interventions, where immersion and interaction are central to the therapeutic experience, emotional responses become even more relevant because they shape how survivors perceive virtual tasks, social participation, and feedback mechanisms. Hence, integrating emotional awareness into rehabilitation represents an important step toward more responsive and human-centered interventions, enabling healthcare professionals to better understand how survivors experience therapy.

This work proposes EMOCAP, a visualization framework for shared emotional awareness in personalized VR for stroke recovery, where emotional information is treated as dynamic data requiring real-time interpretation by distinct actors. The framework integrates (client-side) emotion recognition, a (server-based) distribution layer for broadcasting affective states across connected environments, and an in-situ VR visualization component that encodes emotions through lightweight symbolic representations to support immediate awareness. By externalizing otherwise latent affective signals, the framework can enables stroke survivors, healthcare professionals, and rehabilitation companions to share emotional context during rehabilitation, while establishing visual emotional awareness as a foundational layer for the personalization process of the VR experience. This perspective is especially relevant, since adapting virtual environments requires not only functional performance indicators but also contextual knowledge about engagement throughout the recovery process.

Moving forward, we plan to integrate through other equipment, as is the case of the VIVE XR Elite headset, being capable of presenting the VR stimuli while capturing affective signals such as facial expressions and gaze direction. Afterward, we will validate the framework with stroke survivors and healthcare professionals, ensuring it is aligned with their expectations, and rehabilitation goals. Particular attention will be given to the visualization component, assessing whether the emotional representations are intuitive, minimally intrusive, and capable of supporting shared awareness during rehabilitation. Additionally, we also plan to integrate other modalities, such as sentiment analysis, allowing the framework to incorporate verbal and textual indicators alongside physiological and behavioral emotional cues. This multimodal expansion will strengthen affective interpretation by combining complementary sources of information, while also enriching the visualization layer through more rich and context-aware representations of emotional state.

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